

Cycling Cadence and Why It Matters for Training Design

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Cadence is how many times per minute the pedals turn (RPM). It is one of the most overlooked levers in cycling, because it changes the physiological cost of the same power output.

Same Watts, Different Cost

Take 200 watts as an example:

- **At 70 RPM**, the effort is muscular. Heart rate stays lower, but fatigue rate is higher, and over time the athlete builds extra quad mass. That mass adds weight and generates heat, which is a real liability in hot races.
- **At 90 to 95 RPM**, the effort shifts to the cardiovascular system, where well-trained triathletes already have capacity to spare.

Many athletes never get their cadence up, so a big aerobic engine goes underused while fatigue and muscle mass climb. Getting cadence right lets athletes use the fitness they already have.

General Cadence Targets

Athlete Group	Suggested Cadence	Notes
Long course age group	85 to 90 RPM	Below this is muscularly fatiguing, above it adds neurological tiring
World-class long course	95 to 100 RPM	Higher power at speed needs higher cadence to limit fatigue
Short course	95 to 105 RPM	Higher turnover suits surge-heavy racing
Track cycling background	Up to about 160 RPM	Fixed-gear training builds high-cadence efficiency

Figure 1. General cadence guidance by athlete group. These are starting points, not rules. Some athletes ride very well in the low 80s, so don't force one number on everyone.

Build a Wide Usable Range

Target cadence matters, but so does range. A short course athlete should be efficient from roughly 70 to 110 RPM without constantly changing gears. Constant shifting is a tell that an athlete can't find a gear that feels natural. Holding power across a wide cadence range is a neuromuscular skill worth training.

A Protocol for Raising Natural Cadence

1. During quality intervals, let the athlete ride at the cadence where they produce the most power. Don't force it here.
2. During every warm-up, cool-down, and recovery gap between intervals, hold about 95 RPM.
3. Repeat over months. The cadence the athlete defaults to climbs on its own.

Expect pushback. Athletes usually say the intervals feel fine but the "easy" gaps are the hard part. That discomfort passes.

CASE EXAMPLE: SIX-MONTH CADENCE SHIFT

A top under-23 athlete (about 116 lb) started with a natural cadence of 75 RPM and 165 watts at FTP. Using the recovery-gap protocol at 100 RPM, her habitual cadence moved to 95 to 100 RPM within six months, and her power climbed from 165 to 225 watts with no change in body weight. Raising cadence unlocked aerobic capacity she already had but couldn't use.

The same pattern shows up at the elite level. Chrissie Wellington rode around 75 RPM when she first won the Ironman world title. By the end of her career her cadence was above 90 RPM, which let her carry less muscle mass, keep fatigue down, and run faster off the bike.

How Cadence Fits Into Program Design

- **Read it as a component.** Cadence distribution sits alongside variability index, FTP, FRC, and fatigue rate when you profile an athlete.
- **Set targets by race format.** Short course and long course demand different cadence ranges, so build toward the format the athlete is racing.
- **Use recovery gaps.** Cadence work doesn't need its own session. It rides along inside the sessions you already have.
- **Watch trainer versus road.** The trainer gives precise control of power and cadence, but the road is where race-specific demands show up.
- **Use heart rate as a proxy.** If an athlete has no outdoor power meter, correlate heart rate to power at a steady cadence indoors (for example, 200 watts at 85 to 90 RPM giving a heart rate of 160). That heart rate then guides outdoor training.

KEY TAKEAWAY

Cadence is a free lever. Raising natural cadence into the 90s lowers fatigue and heat-producing muscle mass, and it opens up aerobic capacity the athlete already owns. Hold about 95 RPM in the recovery gaps and let the habit shift over months.